Section A: Making 2D Shapes in OpenSCAD

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These activities show you how to use the OpenSCAD program to create 2D shapes and directly export them as an SVG. OpenSCAD is an open-source, free computer-aided design (CAD) program. It can create 2D shapes in SVG or DXF format (laser/vinyl cutters and embossers), and 3D shapes for 3D printing. This activity focuses on very basic activities. The activities in the "Programming in OpenSCAD" section discuss OpenSCAD and its features in more depth. We also provide links to more resources about coding in general there.

Objectives

- Learn to create precise geometrical shapes in OpenSCAD
- Learn how to combine shapes in various ways and to translate, rotate and mirror them
- Learn how to export shapes in SVG format

Materials

Download and install OpenSCAD on a computer running MacOS, Windows or Linux (not a tablet, iPad, Chromebook or other mobile device). The program (and good documentation) is available at openscad.org. All these activities assume that OpenSCAD has been installed on your computer.

Other Resources

If you need more help, our book Make: Geometry has more detailed OpenSCAD startup instructions. Along with our other math books and a teacher's guide, it is available in the Maker Shed and other book retailers.

One of the creators of OpenSCAD also collaborated on the book Programming with OpenSCAD: A Beginner's Guide to Coding 3D-Printable Objects, by Justin Gohde and Marius Kintel, available from the publisher at nostarch.com or on Amazon.

If you like to watch videos instead, the OpenSCAD team created tutorial videos linked on their site. They do not have formal video descriptions per se but they talk through what they are doing on the screen as they do it.

- There is an OpenSCAD introduction video about the same level as the beginning of our 3D materials. It is roughly nine and a half minutes long.
- For more depth, there is a YouTube playlist of 27 videos that start with downloading and go through progressively more advanced materials.

Ideas for Carrying Out this Activity

OpenSCAD is accessible with a screen reader. One option is to create a file with the examples we have created here and allow them to compare those with tactile outputs. Then, encourage students to change and combine examples, predict what will happen, and then export their version to SVG for a tactile print.

Alternatives

Teachers who can use a visual interface might find it easier to create models in other drag and drop software (such as Tinkercad, available at tinkercad.com). OpenSCAD's great strength is that it makes it possible to create accurate 2D and 3D models with mathematical relationships. For example, it is pretty easy to draw a rectangle and rotate it 45 degrees.

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